English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, 06.02.2025 Exam

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*Grading formula: 2 plus points earned. Time allowed: 3 hours. Good luck!*

**Problem 1.** (0.75 points) A coin is tossed repeatedly, with probability $p$ of heads and $1-p$ of tails. Let $Y$ be the number of tosses needed to obtain exactly 3 heads.

1. (0.25 points) Find $\mathbb{P}(Y = n)$ and $\mathbb{E}[Y]$.
2. (0.5 points) Express $\mathbb{P}(Y \leq 2k)$ and, for $p = 1/2$ and large $k$, give an approximation.

**Problem 2.** (1 point) Box A contains slips bearing the integers from 1 to 10 inclusive. Angel chooses $K$ uniformly from 1 to 10, draws $K$ randomly chosen slips from box A, and places them in box B.

1. (0.25 points) What is the expected number on a randomly drawn slip from box B?
2. (0.25 points) What is the probability that box B contains two numbers whose sum is 11?
3. (0.5 points) Let $X$ be the smallest number in box B, and let $N$ be the number of draws with replacement from B until $X$ is drawn for the first time. Is $N$ geometrically distributed, and if so, with what parameter? Find $\mathbb{E}[N]$.

**Problem 3.** (1.25 points) Let $X, Y, Z$ be random variables with respective densities

$$f_X(x) = \begin{cases} k\,x\,e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases} \qquad \text{and} \qquad f_{Y,Z}(x,y) = \begin{cases} c\,(x+y) & x > 0,\ y > 0,\ x+y < 1 \\ 0 & \text{otherwise} \end{cases}$$

where $k, c$ are constants and $X$ is independent of $(Y, Z)$.

1. (0.25 points) Find $k, c$ and the expectations of $X, Y, Z$.
2. (0.25 points) Find $\mathbb{P}(X > 2)$ and $\mathbb{P}(Y > 1/2)$.
3. (0.5 points) Are $Y$ and $Z$ independent? Find their covariance. Calculate $\mathbb{E}[Z|Y]$, $\mathbb{E}[X|Y]$ and $\mathbb{E}[Z|(Y = \mathbb{E}[Y])]$.
4. (0.25 points) Find the density of $\eta := YZ$.

**Problem 4.** (1.0 point) Let $\sigma$ be a uniformly chosen permutation of $\{1, 2, \ldots, n\}$ among all $n!$ possibilities. Recall that a permutation of $\{1, 2, \ldots, n\}$ is a bijection

$$\sigma : \{1, 2, \ldots, n\} \to \{1, 2, \ldots, n\},$$

that assigns each number from 1 to $n$ exactly one image and conversely. Every permutation decomposes into disjoint sets called cycles. A cycle of length $k$ is a sequence of distinct elements $(a_1, a_2, \ldots, a_k)$ satisfying:

$$\sigma(a_1) = a_2, \quad \sigma(a_2) = a_3, \quad \ldots, \quad \sigma(a_{k-1}) = a_k, \quad \sigma(a_k) = a_1.$$

Define the random variables

$$c(\sigma) := \text{number of cycles in } \sigma, \quad F(\sigma) := \text{number of fixed points, i.e. } i \text{ such that } \sigma(i) = i.$$

1. (0.25 points) Show that, for every $1 \leq k \leq n$, the probability that 1 belongs to a cycle of length $k$ is $1/n$.
2. (0.25 points) Find $\mathbb{E}[c(\sigma)]$ and $\mathbb{E}[F(\sigma)]$.
3. (0.25 points) Find the probability that $\sigma$ has no fixed points, and its limit as $n \to \infty$.
4. (0.25 points) Is it true that
$$\frac{c(\sigma) - \log n}{\sqrt{\log n}} \xrightarrow[n\to\infty]{d} N(0,1)?$$
Justify your answer fully, stating the assumptions and theorems needed and performing the relevant calculations.